

# **Institute of Actuaries of India**

## **Subject SP8 – General Insurance: Pricing**

### **March 2022 Examination**

## **INDICATIVE SOLUTION**

#### **Introduction**

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

**Solution 1:****i) Assumptions:**

- Each group of the districts contain the same amount of past exposure in each allocation. (1)
  - Data considered belongs to exponential family. (0.5)
  - There is a linear predictor and link function. (0.5)
- (2)**

**ii) Response variable**

$$Y_i = g^{-1}(X_i \beta + \xi) + \varepsilon \quad (1.5)$$

Where

$g()$  is the link function (0.5)

$X$  is the design matrix of factors (0.5)

$\beta$  is a vector of parameters to be estimated (0.5)

$\xi$  is a vector of offsets or known effects (0.5)

$\varepsilon$  is the error term appropriate to  $Y$  (0.5)

**iii)**

- The most important consideration is the degree of fit to the expected future experience.
- Degree of fit can be tested using formal statistical tests in the modelling process. e.g. Akaike Information Criterion, or other valid test (not chi-square, as models are not nested).
- There might be a rapid trend over time towards either A or B.
- A time consistency check involves interacting the district with a time-related factor and looking for a trend.
- Another check is to subdivide by a random factor.
- The degree of uncertainty of the model parameters can be assessed by calculating standard errors of the parameter estimators.
- The spread of relativity values across RTO codes can be combined with their standard errors to check that their error ranges do not overlap too much.
- Over-fitting is a danger because this causes the model to lose predictive power.
- A check on this is to withhold some data from the sample used to fit the model and perform tests on the model's fit to the withheld data (or use an out-of-time dataset).
- Cook's distance can be used to see whether any data points have an undue influence over the choice of A or B. e.g. young drivers in urban areas, or a large liability claim.
- Model A's graph is steeper than B's, so A differentiates more between good and poor experience for this factor.
- However, more discrimination between risks is only helpful if the fit is also better.
- A lift curve could be constructed to compare the predictive power of A and B on an out-of-sample dataset.
- One method is to rank all out-of-sample data by expected burning cost for each model separately, then plot a graph of actual experience against each of those rankings on the same chart, using the same exposure scale.
- The steeper the curve, the more effective the model is at distinguishing high from low burning cost.
- A gains curve can also give information on the value added by the district classification.
- With this method the data is sorted high to low according to the fitted model values and a graph can be plotted to show the cumulative values of the fitted model and the observed values from the data against cumulative exposure. This could be done for each of A and B.
- The Gini coefficient can then be calculated to provide a statistical measure for the lift

produced by the model. This can be thought of as the area enclosed by the model curve and the diagonal line as a ratio of the triangle above the diagonal.

- Curve A is not as smooth as curve B, so we may wish to smooth/adjust it before using it. We should beware of smoothing too much, as this will reduce fit.
- Curve A slopes the wrong way for few RTO codes, so this may need to be looked into further and adjusted for.
- We would need to consider the market's approach to this factor, i.e. do some market comparisons for various postcodes, because we might not want to be too far out of line for policies in our target market and/or we may not need to be as cheap for postcodes in the low districts.
- Consider how our mix of business might change in the future depending on which of the two RTO allocations we choose.
- We would also need to consider the extent of the change from the existing district structure, e.g. by plotting it on the same graph, for comparison, and particularly the impact on customer price, e.g. identify where large swings in prices are expected.
- How practical is it to implement either structure (ie, is one more complicated than the other)?
- Will either structure result in large jumps in premium for a small change in distance (risking customer dissatisfaction)?
- Does the choice of A or B make a significant difference to the overall model?

(0.5 mark for each point , Max 10)

#### iv) Advantages

- The presence of nil claims can distort the average claim size.
- Especially if there is a trend in nil claims.  
...e.g. increased fraud detection increases probability of claim being rejected.  
...or above-inflation increases in excess.
- This could lead to incorrect pricing
- Excluding nil claims can make it easier to compare results with competitors or industry benchmarks
- May take some time to reject a claim, so may be very few recorded nil claims in more recent experience
- Having to adjust for this will be subjective/difficult.
- Easier to fit distribution when point mass at zero is removed e.g. gamma for claim severity

#### Disadvantages

- There is no standard definition for a nil claim so may not be easy/quick to do and the definition could also vary over time, adding to the complexity.
- ...and making comparisons to industry benchmarks difficult
- If nil claims are more prevalent in certain classes of business e.g. Liability then ignoring them means valid trends could be missed
- Important to retain cases where the claim is below the excess so that insurer can model potential reductions in excess – these would now be non-nil (i.e. we want to know the distribution of losses below the excess).

(Max 4)

**[20 Marks]**

#### Solution 2:

- i) The Banks may experience the following losses due to external non-physical attacks on bank's data and systems:

- Business disruption due to technology failures or outages i.e. business interruption, contingent business interruption and systems failure -
- Expense and P&L loss resulting from a breach of security at Bank or vendors and suppliers (e.g. exchanges) or from a simple failure of Bank's trading/technology platforms.

- Information / data leakage
- Loss of client or proprietary financial assets (i.e. fraud loss)
- Judgments, civil awards, or settlements after data breach.
- Misappropriation of client's financial assets (e.g. money and securities) in the Bank's care custody and control through phishing, malware, email hijacking and other tactics.
- Regulatory fines
- Reputational damage
- Electronic media liability, including infringement of copyright, domain name, trade name, service mark, or slogan on an intranet or Internet site
- Losses due to a threat of extortion, professional fees related to dealing with the extortion
- Other relevant losses

(0.5 mark for each other point; Max 4)

ii) The Company could face the following risks while introducing the new products:

- Data challenges
  - The Company could lack data
  - ...or understanding of the risk,
  - ...thereby impacting the price charged for risks.
  - Additionally, the underlying rating factors might not be known completely which could lead to cross-subsidies across different risks.
  - Could also result in anti-selection as the cover is not well known in the market
  - The Company also lacks claims development data to undertake adequate reserving
  - . . . leading to financial loss
- Loose policy wording would mean that the coverage provided could be much wider than originally envisaged and priced for
- . . . or additional claims disputes
- Lack of relevant operational capability
- . . . or adequate pre-launch testing impacting
  - Claims handling
  - IT
  - Sales
  - Customer service
- Wrong assessment of demand levels
  - Too low to cover fixed expenses
  - Too high to manage effectively
- Concentrations in segments indicating anti-selection
- Slow to market so lose first mover advantage
- Competition higher than expected
- Liquidity issues
- Reputational impacts
- Regulatory issues
- Reinsurance issues

(0.5 mark for each other point; Max 5)

**[9 Marks]**

### **Solution 3:**

i) Risk premium calculation:

Expected pure risk premium =  $5\% \times 20000 = 1000$  (0.5)

Expected number of claims =  $5\% \times 1,20,000 = 6000$  (0.5)

$\Phi(y) = [\frac{1}{2}] \times (1 + 80\%) = 0.90$  (0.5)

$y = 1.282$  (0.5)

Number of claims for full credibility

$$N_S = \left( \frac{y}{k} \right)^2 \left( \frac{\sigma_N^2}{\mu_N} + \frac{\sigma_X^2}{\mu_X^2} \right) \quad (1)$$

Here  $k = 0.1$

$$N_S = (1.282/0.1)^2 * ((1-.05)+1) = 321 \quad (0.5)$$

Credibility factor  $Z = (6000 / 321)$

$$\square \text{ Hence } N_S = 2(1.645/0.1)^2 = 541.205$$

As the number of claims is more than credible number of claims, credibility factor to be associated with the experience is 1.

Hence the credibility weighted premium is Rs 1000 per vehicle year. (0.5)  
**(4)**

**ii) Assumptions:**

- claim frequency and claim severity are independent of each other
- Estimated number of claims for IBNR has been included in the actual number of claims reported
- Any open claim at the time of submitting the data has been assumed to be settled at the same amount
- There will be no change in the mix of vehicles in the next coverage year
- There is no loading for unusually large claims or a catastrophe
- No inflation adjustment is needed on the past year risk premium
- No changes in T&C/underwriting/claims handling
- No changes in the external claims environment

(0.5 mark for each point , Max 3)

**iii) Other data required for Bayesian credibility:**

- Distribution of claims,  $f(x/\theta)$  is required, from which  $E(x/\theta)$  and  $\text{var}(x/\theta)$  can be calculated. (2)
- Prior distribution  $\Pi(\theta)$  needs to be defined. (1)

**(3)**

**[10 Marks]**

**Solution 4:**

- It is helpful to be aware of competitive position because it helps to estimate impact of price changes on volumes and income.
- The method could work well if the product has few rating factors.
- Tracking the market may be useful if some of the individual products have only small volume, or past data is unavailable or the insurer is totally new to this product offering.
- However, this “large” book should have good enough volume of data for using own experience.
- Failure to use the company’s own experience may result in a higher capital requirement, or a higher reinsurance cost.
- There could still be enough scope for variation between the company’s prices and the competitor’s, even if the average is similar.
- It might be better to use more than one competitor to avoid large price swings.
- The constraint on pricing strategy could erode profitability overall (ie be sub-optimal) in the following ways:
  - Writing some risks at unprofitable rates.

- Losing business by charging too much for some risks.
- The price comparison may be distorted/invalidated by:
  - Product features and benefits not being identical.
  - Periods of time in which the competitor is running a special promotion.
  - Insufficient volumes in the basket of risks.
  - A small number of very high prices skewing the average.
  - Differences between insurers, such as expense base, reinsurance structure.
  - Strategy of the competitor, such as loss leaders or desire for growth.
- If open-market prices are readily available then it could be a quick and easy method of setting a price.
- However, it could be time-consuming and expensive because telephone and face-to-face channels must be worked manually, and internet sites may have anti-screen-scraping measures.
- Some prices may not be visible at all, for example negotiated discounts, affinity or loyalty discounts.
- If the company wants to apply the approach for renewals as well as new business, reliable renewal prices will be almost impossible to obtain.
- There may be legal issues with obtaining the data, such as legislation on accessing websites, mystery shopping or competition law.
- Prices could become out-of date very quickly.
- If the two insurers represent a large share of the market, this practice could amplify the insurance cycle.
- The practice could become known to the public, which could erode consumers' confidence in the industry or give rise to an investigation by regulator.

**[0.5 mark for each point, Max 5 Marks]**

#### **Solution 5:**

- Quick and easy to calculate (0.5)
- Unlikely to be appropriate as there is no information on exposure/size of fleet (1)
- if exposure is growing or shrinking, would need to restate historic experience accordingly (0.5)
- 5 years is unlikely to be sufficient experience for these types of risks (0.5)
- in particular, some potential risk events may not have happened in previous 5 years, but should allow for non-zero probability of such risks in pricing (0.5)
- Likewise the observed losses may not reflect the true underlying risks, as the period over which losses have been observed may be much shorter than the return period of the losses under consideration. (0.5)
- cat losses appear to be trending upwards, which straight-average doesn't properly capture; would need to understand reasons for this (0.5)
- increase may be due to changes in mix of business (0.5)
- or just very prudent case reserves in recent years (0.5)
- no allowance has been made for inflation/trends (0.5)
- the incurred claims may need to be projected to ultimate depending on how they have been defined (0.5)
- It is not clear whether the incurred losses are gross or net of reinsurance – any year-on-year changes in reinsurance need to be allowed for (0.5)
- As it uses data provided by the insured it may have errors or omissions (0.5)
- Does not allow for changes in the external environment including exchange rates (0.5)
- Nor for any possible changes in legislation which may affect the number and cost of claims for catastrophes (0.5)
- Although the book may be large it may still not be appropriate to base the loading entirely on the own experience (i.e. consider credibility). We may want to look at catastrophe loadings for similar risks we insure, or some other form of benchmark cat loading percentage. (0.5)
- Cat models may be used (0.5)

**[Max 5 Marks]**

**Solution 6:****i) Information could be entered onto the wrong claim record.**

The claim and policy numbers should both be in series that mistakes are difficult to make — a single digit wrong or two numbers swapped will give an invalid number, and it should not be possible for a policy number also to be a claim number.

**Incorrect amounts may be entered, or correct amounts in the wrong currency.**

There should be some check on amounts. Very large or small claims should be queried if entered. This is especially important if working in a variety of currencies. A query should be raised if an amount is entered in a different currency from previous entries. A query should be raised if the claim is not in the currency of the country of the address of the policyholder (this will not apply for marine insurance, travel insurance and some other classes).

**A claim may be entered for an incorrect date.**

The system should automatically check that the policy was on-risk on the day when the claim occurred. A query should be raised if there is a very long gap between the date of loss (or reporting to the insured) and reporting to the insurer, or if the date of loss was later than the date reported.

**Information may be entered against the wrong policy.**

Other details, such as policyholder surname, deductible, and the fact that paid + estimated outstanding < sum insured, should be checked against the information on the policy record.

**Information may be missed out.**

A claim should not be accepted until all fields have been filled in, possibly with null entries.

(5)

**ii)**

- **Unusually heavy or light experience:**

- Claims experience tends to go in cycles
- For some classes unusually heavy or light years may be experienced in isolation especially if the risk is affected by climate
- If experience is untypical then choose another base year or aggregate more years' experience or apply an adjustment factor to the base year which would obviously be subjective although industry data may be available

- **Large or exceptional claims:**

- May be left in the data
- Or truncated and spread
- Or removed
- Depending on the extent to which similar claims are likely to occur in the future

- **Trends in claims experience**

- If trends are detected in the base data, it is important to attach more weight to recent experience
- Allowance for inflation
- Trends should also be investigated to see whether or not they are likely to continue into the future
- Or if they are the results of a one-off change in company or market practice.
- If they are expected to continue then an assumption will be needed to allow for them.

- It may be necessary to adjust past data
- **Changes in risk:**
  - Changes in risk can be difficult to deal with.
  - They may show up as trends and be dealt with as such.
  - Alternatively, major elements of the risk could be separated in the base data and projected separately and combined with an assumption about the future mix of risks.
- **Changes in cover:**
  - Changes in cover can be difficult to allow for.
  - Major changes are likely to involve the perils covered Or the limits and excesses applied to each claim.
  - They may also arise from changes to underwriting or to claims settlement procedures.
  - If a peril is no longer to be insured, it may be possible to exclude these claims from the data.
  - If a new peril is to be insured, it may be necessary to use external data such as market statistics, consumer or manufacturer data, government statistics.
  - Changes to limits or excesses are more complicated.
  - If there is a detailed database allowing all claims to be separately considered, it may be possible to adjust each claim to the original gross amount and project the gross data to the new rating period.
  - Otherwise, it will be necessary to make more approximate adjustments based on any knowledge of the underlying claims cost distribution.
  - Either way the information will be incomplete, as many insureds will not notify claims below or near the excess points.
  - Future changes in the risk environment other than normal trends will need to be identified.
- **Changes in reinsurance cost:**
  - It will be necessary to allow for changes in reinsurance cost.
  - May be necessary to incorporate IBNR
  - Margins in reserves (positive and negative)
  - Errors in data
  - Changes in claim definition including treatment of NIL cost claims

(0.5 mark for each point, Max 12)

iii)

- **Failure of third parties for example**
  - Nonpayment from brokers
  - Staff dishonesty
  - Default by supplier
  - Delay in premiums, Delay in recoveries
- **Competition**
  - Products are inferior
  - Prices uncompetitive, high expenses, inadequate analysis
  - Fall behind in technology, less business
  - Prices too cheap, unsatisfactory financial performance
  - Business affected by changes is legislation which other have allowed for
  - If less sales, business costs rise as fixed expenses have to be recouped
  - Data may be inaccurate / incomplete
- **Insurance cycle**
  - As the insurer only writes motor business there is no opportunity to cross subsidies with classes at different stages of the cycle.
  - As profits increase insurers enter the market, buy market share with low premiums, market premiums decrease until losses are made. Insurers leave the market; prices increase and the market edges back into profitability.

- As only write one class at the bottom of the cycle will either lose business (puts pressure on fixed expenses) or lower premiums and decrease the solvency position.
- **Other business risks include expenses, investment and management incompetence.**  
(0.5 mark for each point; Max 4)

iv)

- Excess of Loss protection and Catastrophe XOL for large single events
- Surplus reinsurance for ceding out larger risks.
- Stop Loss / Aggregate XL reinsurance.
- Exclusions (Terrorism / NCB)
- Limits on size of risks that can be accepted.
- Reduce exposure to certain industries / types of businesses that exhibit greater variability of claim experience (e.g. Oil industry or large multinationals)
- Track and control aggregates by geographic region.
- Diversify by region, industry, type of business.
- Match liabilities and assets by currency in order to reduce currency risk
- Restrict the territories / currencies that it writes business in to reduce currency risk.
- Place reinsurance with financially strong reinsurers (AAA rated)
- Spread the reinsurance placement amongst several reinsurers.
- Employ specialist wordings staff to reduce risk from ambiguous policy wording.
- Impose excess / deductibles
- Impose catastrophe / event limits or other sub-limits
- Stricter claims underwriting to prevent fraudulent claims
- Review rating structure and rates more frequently to reduce anti-selection
- Diversify into other areas of insurance

(0.5 marks for each correct point, Max 7)

**[28 Marks]****Solution 7:**

i)

| Year | Exposure | Mid year Exposure | Paid claims  | % developed | Ultimate     | Inflation for Risk premium - 2022 | Inflation for Risk premium - 2023 |
|------|----------|-------------------|--------------|-------------|--------------|-----------------------------------|-----------------------------------|
| 2016 | 1200     | 1212.5            | 5,00,00,000  | 99%         | 5,05,05,051  | 1.340095641                       | 1.407100423                       |
| 2017 | 1225     | 1266.5            | 6,75,00,000  | 97.50%      | 6,92,30,769  | 1.276281563                       | 1.340095641                       |
| 2018 | 1308     | 1453              | 9,20,00,000  | 92%         | 10,00,00,000 | 1.21550625                        | 1.276281563                       |
| 2019 | 1598     | 1349              | 12,15,00,000 | 85%         | 14,29,41,176 | 1.157625                          | 1.21550625                        |
| 2020 | 1100     | 1100              | 9,40,00,000  | 79%         | 11,89,87,342 | 1.1025                            | 1.157625                          |

| Ultimate claims -2022 after inflation | Ultimate claims -2023 after inflation | Annual BC - 2022 | Annual BC - 2023 |
|---------------------------------------|---------------------------------------|------------------|------------------|
| 6,76,81,598.01                        | 7,10,65,677.91                        | 55819.87465      | 59221.39826      |
| 8,83,57,954.33                        | 9,27,75,852.04                        | 69765.4594       | 75735.38942      |
| 12,15,50,625.00                       | 12,76,28,156.25                       | 83654.93806      | 97575.043        |
| 16,54,72,279.41                       | 17,37,45,893.38                       | 122662.9202      | 108727.0922      |
| 13,11,83,544.30                       | 13,77,42,721.52                       | 119257.7675      | 125220.6559      |
| Average BC                            |                                       | 90232.19198      | 93295.91577      |

Assumptions:

Claims are made evenly through the year

Exposure changes evenly through the year

Inflation continues at 5% p.a.

No change in policy terms over the period

The development %s given are for paid claims, not reported incurred

(7)

ii)

- Existence and amount of large/catastrophe claims
- Useful to have a breakdown of the claims by peril/heads of damage
- Information about changes the insured may have made that could affect the risk e.g. Experience of drivers, Nature of items delivered, Changes in risk management
- Territories or countries covered and claims and exposure split by these
- Information about any changes in T&Cs/excesses/limits in the past
- and how these might differ in the future period of cover
- Whether the same insurer has covered the fleet over the past 5 years, as different insurers could have different T&Cs, and different levels of scrutiny in underwriting and claims handling
- Regulatory or legislative changes that may have impacted past experience
- or likely to impact future experience
- Given the large year-on-year increase in burning cost would want to thoroughly check data and assumptions and compare with similar books of business and/or industry data, including for example changes in economic conditions over time
- and gaining insight from experienced underwriters in the market

(0.5 mark for each point, Max 3)

**[10 Marks]**

### **Solution 8:**

#### **i) Benefits/Advantage**

- It enables an insurer to write larger risks, which might otherwise be beyond its writing capacity.
- It enables the insurer to choose, within limits, the size of risks that it will retain.
- Can choose to cede which risks you want, which isn't possible with obligatory QS
- It is better for those classes where a wide variation can occur in the size of risks.
- As a result of all of the above, it can help the cedant to achieve a better portfolio balance

#### **Limitations/Disadvantages**

- The administration is more complicated than for quota share owing to the need to assess and record separately for each risk the amount to be ceded.
- This makes it unsuitable for mass-market personal lines.
- The size of risks is too small to merit individual attention.
- The choice of reinsurers or terms offered may not be as favorable as QS because of the possible anti-selection risk borne by the reinsurer.
- If the facility is facultative-obligatory, the direct writer may forget to cede a large risk.

(4)

#### **ii) Profit Commission**

The Profit Commission could be seen as an incentive from the reinsurer to the Ceding Company that consists in paying back a part of the Treaty profitability.

This is to encourage the Ceding Company to a quality underwriting and good management of the Agreement to optimize its profits.

(2)

iii) Expense Allowance =  $140,000 \times 0.15 = 21,000$

$$\text{Profit Commission} = (140,000 - 21,000 - 16,000) \times 0.30 = 30,900$$

(3)

iv)

| Max lines allowed 6 |                        |           |                  |               |       |               |               |                |
|---------------------|------------------------|-----------|------------------|---------------|-------|---------------|---------------|----------------|
|                     | 1                      | 2         | 3                | 4             | 5     | 6             | 7             | 8              |
| Risk                | Estimated Maximum Loss | Retention | Original premium | Gross loss    | LOC   | ceded (5*2/1) | Ced Pre (3*6) | Ced loss (6*4) |
| 1                   | 5,000,000              | 1,000,000 | 38,000           | 20,000        | 4     | 80%           | 30,400        | 16,000         |
| 2                   | 10,000,000             | 800,000   | 75,000           | 25,000        | 6     | 48%           | 36,000        | 12,000         |
| Total               |                        |           | <b>113,000</b>   | <b>45,000</b> |       |               | <b>66,400</b> | <b>28,000</b>  |
|                     | GLR                    |           | 113,000          | 45,000        | 39.8% |               |               |                |
|                     | CLR                    |           | 66,400           | 28,000        | 42.2% |               |               |                |

Below marks for correct calculation:

GLR (1)

correct ceded % (1)

correct Ceded prem and loss (1)

correct Ceded LR (1)

(4)

**[13 Marks]**

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